Geonu Lee

AI Research Engineer

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Work Experiences

Jul. 2024 - Present

Full-time – SNUAILAB

AI Research Engineer

- **Led the end-to-end development and deployment** of an AI-based smartphone cover glass micro-defect inspection system, achieving a False Negative Rate (FNR) below 2%.
- **Prototyped CAD-to-scan point cloud registration**, extracting reference point clouds from CAD models and aligning them to 3D scans of manufactured parts for geometry-based inspection.
- **Built a Blender-based virtual defect generation tool and its client-facing GUI**, mirroring the production imaging setup to synthesize defect images for manufacturing inspection.
- **Managed technical alignment and crisis resolution** with a manufacturing client, establishing technical trust that successfully secured strategic corporate investment.
- **Trained deep learning models** for a battery pole tracking system and integrated real-time pipelines into a C++ SDK, stabilizing edge deployment for a specialized inspection equipment manufacturer.
- **First-authored** "Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection", *Neurocomputing*.

Oct. 2023 - Apr. 2024

Full-time – ALCHERA

AI Research Engineer, Anomaly Analysis Team

- **Developed and deployed DETR-based object detection models** for FireScout, a real-time vision-based wildfire detection solution, achieving a 20%p performance improvement over the legacy baseline.
- **Reconfigured the evaluation pipeline** from image-level to instance-level metrics, resolving legacy evaluation limitations to establish a precise and robust performance measurement system.

Mar. 2023 - Sep. 2023

Intern – Naver Cloud

AI Research Engineer, Image Vision Team

- **Contributed to the development of a face anti-spoofing service framework** for eKYC services, constructing a robust system capable of detecting spoofing attacks across diverse environmental conditions.
- **First-authored a paper published at WACV 2025** on object anti-spoofing, improving HTER by 7.5% and EER by 6.6% to significantly enhance domain generalization performance.

Jan. 2020 - Feb. 2020

Intern – Electronics and Telecommunications Research Institute (ETRI)

AI Research Engineer

- Developed a human action recognition system that models interactions between humans and surrounding objects in video data.

Research Interests

3D Representation Learning, Domain Generalization, Continual Learning, Anomaly Detection, Point Cloud Registration, Embodied AI, World Models, Vision-Language-Action

Publications

Jun. 2026 | **Geonu Lee**, Yujeong Oh, Geonhui Jang, Soyoung Lee, Jeonghyo Song, Sungmin Cha, YoungJoon Yoo, "Continual-MEGA: A Large-scale Benchmark for Generalizable Continual Anomaly Detection," in *Neurocomputing*

Feb. 2025 | **Geonu Lee**, Yonghyun Jeong, Haneol Jang, YoungJoon Yoo, "Domain-Generalized Object Anti-Spoofing: Bridging Gaps and Patch Selection for Robust Detection across Domains," in *Winter Conference on Applications of Computer Vision (WACV)*

Sep. 2022 | **Geonu Lee**, Kimin Yun, Jungchan Cho, "Occluded Pedestrian-Attribute Recognition for Video Sensors Using Group Sparsity," in *Sensors*, vol. 22, no. 17, pp. 6626

Aug. 2022 | **Geonu Lee** and Jungchan Cho, "STDP-Net: Improved Pedestrian Attribute Recognition Using Swin Transformer and Semantic Self-Attention," in *IEEE ACCESS*, vol. 10, no.1, pp. 82656 - 82667

Feb. 2021 | **Geonu Lee**, Kimin Yun, and Jungchan Cho, "Improved Human-Object Interaction Detection through On-the-Fly Stacked Generalization," in *IEEE ACCESS*, vol. 9, no. 1, pp. 34251-34263

Feb. 2020 | Bhishan Bhandari, **Geonu Lee**, and Jungchan Cho, "Body-Part-Aware and Multitask-Aware Single-Image-Based Action Recognition," in *Applied Science*, vol. 10, no. 4, pp. 1531-1548

Under Review

Jul. 2026 | **Geonu Lee**, Kimin Yun, YoungJoon Yoo, "Learning Canonical Representations for Rotation-Invariant and Unified 3D Anomaly Detection," under review

Research Projects

2022 | **A Study on the Complex Human Attributes for Situation Understanding**
Master's Research Project – Gachon University
- Conducted research on recognizing interactive human attributes in video sequences under complex contexts.
- Utilized Swin Transformer and Transformer decoder architectures for semantic reasoning.
- Funded by Electronics and Telecommunications Research Institute (ETRI).

2021 | **A Study on the Understanding of Pedestrians**
Master's Research Project – Gachon University
- Developed a pedestrian attribute recognition method robust to occlusion.
- Applied group sparsity regularization to handle various levels of visual obstruction.
- Funded by Electronics and Telecommunications Research Institute (ETRI).

2020 | **A Study on the Understanding of Human-Object-Interactions**
Undergraduate Research Project – Gachon University
- Designed a novel deep neural architecture based on on-the-fly stacked generalization for HOI detection.
- Focused on modeling dynamic interactions between humans and surrounding objects.
- Funded by Electronics and Telecommunications Research Institute (ETRI).

2019 | **A Study on the Understanding of Human Situation Based on Deep Learning**
Undergraduate Research Project – Gachon University
- Proposed a multi-task learning framework combining human pose estimation and action recognition.
- Addressed real-world situational understanding using contextual cues.
- Funded by Electronics and Telecommunications Research Institute (ETRI).

Education

Mar. 2021 - Feb. 2023 | **Gachon University**
M.S. in Software Engineering

Mar. 2016 - Feb. 2021 | **Gachon University**
B.S. in Department of Computer Engineering

Academic Service

2026 | Reviewer, International Conference on Machine Learning (ICML)